

Github Tasks

Task 1: Install git.

Objective:

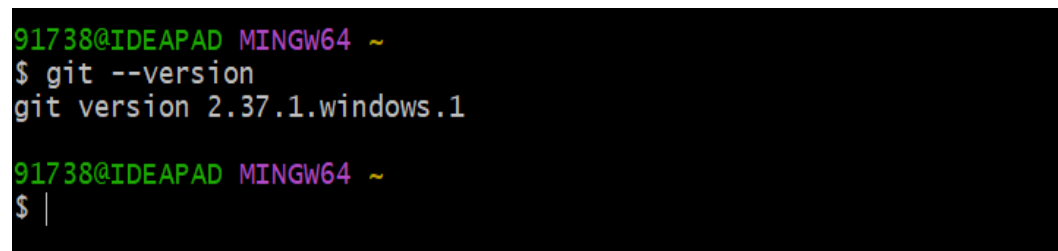
Install Git on the system.

Purpose:

To set up Git for version control and GitHub operations.

Implementation:

1. Installed Git using the package manager.
2. Verified the installation using `git --version`.
3. Confirmed the installed Git version.

A terminal window with a black background and green text. The prompt is '91738@IDEAPAD MINGW64 ~'. The user enters '\$ git --version' and the output is 'git version 2.37.1.windows.1'. The prompt is then '\$ |' with a cursor.

```
91738@IDEAPAD MINGW64 ~  
$ git --version  
git version 2.37.1.windows.1  
  
91738@IDEAPAD MINGW64 ~  
$ |
```

Conclusion:

Successfully installed and verified Git.

Task 2: Create a repo in github with README.md and .ignore file.

Objective:

Create a new GitHub repository with basic project files.

Purpose:

To initialize a repository with a README file and configure files that should not be tracked by Git.

Implementation:

1. Logged in to GitHub.
2. Created a new repository.
3. Enabled **Add a README file**.
4. Selected a .gitignore template.
5. Created the repository.

6. Verified README.md and .gitignore.

Configuration

Choose visibility *
Choose who can see and commit to this repository

Public

Add README
READMEs can be used as longer descriptions. [About READMEs](#)

On ☒

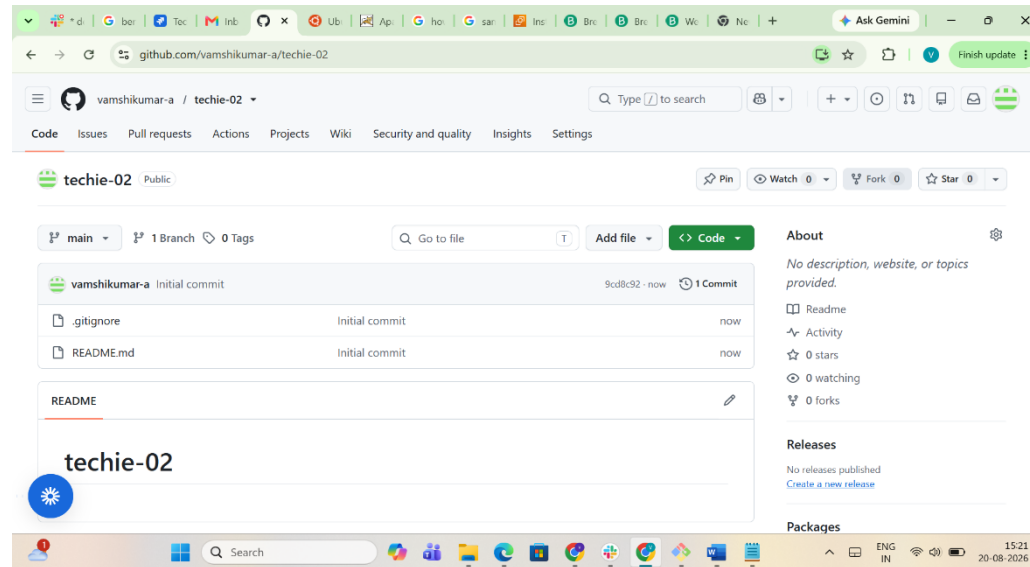
Add .gitignore
.gitignore tells git which files not to track. [About ignoring files](#)

Node

Add license
Licenses explain how others can use your code. [About licenses](#)

No license

Create repository



Conclusion:

Successfully created a GitHub repository with README.md and .gitignore.

Task 3: Clone the created repo to local.

Objective:

Clone the GitHub repository to the local system.

Purpose:

To create a local working copy of the remote repository.

Implementation:

1. Copied the repository URL from GitHub.
2. Used the git clone command.
3. Navigated to the cloned repository.

4. Verified the repository files.

```
91738@IDEAPAD MINGW64 ~/Desktop
$ git clone git@github.com:vamshikumar-a/techie-02.git
Cloning into 'techie-02'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.

91738@IDEAPAD MINGW64 ~/Desktop
$
```

Conclusion:

Successfully cloned the GitHub repository to the local system.

Task 4: Create two files in local repo.

Objective:

Create two files inside the local Git repository.

Purpose:

To practice creating and tracking new files using Git.

Implementation:

1. Navigated to the local repository.
2. Created two files – touch a b.
3. Checked the files using ls.

```
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ ls
README.md

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ touch a b

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ ls
README.md  a  b

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ ls -a
./  ../  .git/  .gitignore  README.md  a  b
```

Conclusion:

Successfully created and verified two files in the local repository.

Task 5: Commit two files and push to central Repository.

Objective:

Commit the created files and push them to GitHub.

Purpose:

To understand the Git workflow of staging, committing, and pushing changes.

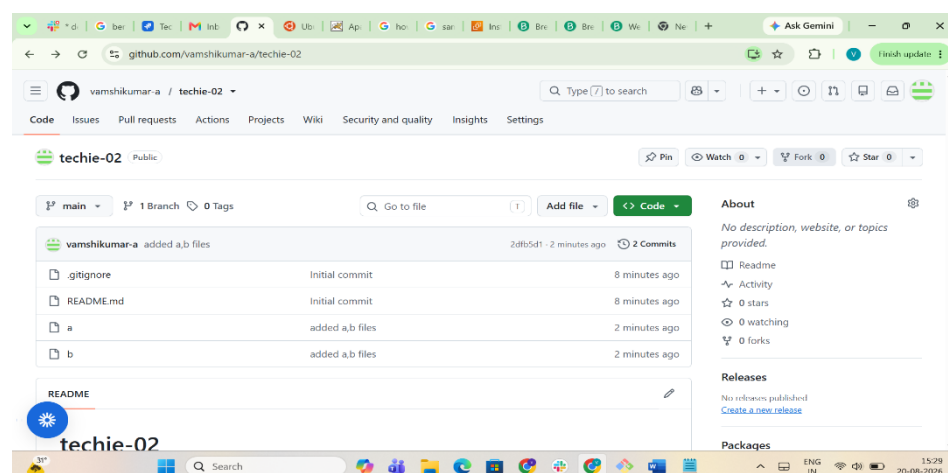
Implementation:

1. Checked the repository status – git status.
2. Added both files using - git add.
3. Created a commit with a meaningful message – git commit -m "message".
4. Pushed the commit to the remote repository - git push -u origin main.
5. Verified the files on GitHub.

```
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        a
        b

nothing added to commit but untracked files present (use "git add" to track)
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ git add .
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ git commit -m "added a,b files"
[main 2dfb5d1] added a,b files
2 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 a
create mode 100644 b
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 311 bytes | 311.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:vamshikumar-a/techie-02.git
   9cd8c92..2dfb5d1  main -> main
branch 'main' set up to track 'origin/main'.
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
```



Conclusion:

Successfully committed and pushed the files to the central repository.

Task 6: Create a branch in local and create a sample file and push to central.

Objective:

Create a new local branch and push its changes to GitHub.

Purpose:

To understand branch creation and isolated development.

Implementation:

1. Created a new local branch – git branch release.
2. Switched to the branch – git checkout release.
3. Created a sample file – touch release.
4. Added and committed the file .
5. Pushed the branch to GitHub.
6. Verified the branch on GitHub.

```
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ git branch release

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ git branch --list
* main
  release

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ git checkout release
Switched to branch 'release'

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (release)
$ touch release

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (release)
$ ls
README.md  a  b  release

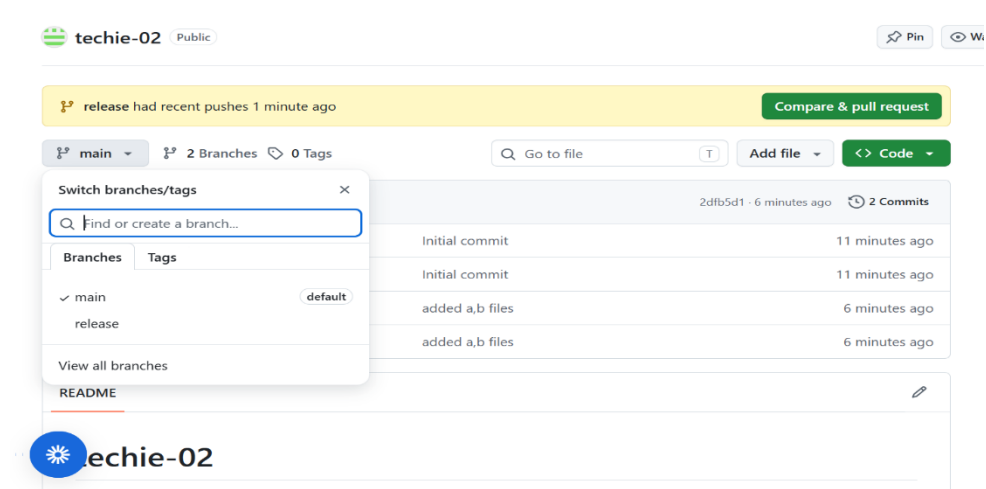
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (release)
$ git status
On branch release
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        release

nothing added to commit but untracked files present (use "git add" to track)

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (release)
$ git add .

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (release)
$ git commit -m "adding release file"
[release 63f33bb] adding release file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 release
```

```
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (release)
$ git push -u origin release
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 245 bytes | 245.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'release' on GitHub by visiting:
remote:   https://github.com/vamshikumar-a/techie-02/pull/new/release
remote:
To github.com:vamshikumar-a/techie-02.git
 * [new branch]      release -> release
branch 'release' set up to track 'origin/release'.
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (release)
$
```



Conclusion:

Successfully created and pushed a local branch to GitHub.

Task 7: Create a branch in github and clone that to local.

Objective:

Create a remote branch and access it locally.

Purpose:

To understand remote branch management and branch tracking.

Implementation:

1. Created a branch from the GitHub repository.
2. Fetched the remote branches locally – git fetch origin.
3. Switched to the branch.
4. git branch –list.

release had recent pushes 1 minute ago

main 2 Branches 0 Tags

Go to file

Switch branches/tags

hotfix

Branches Tags

Create branch hotfix from main

View all branches

Initial commit

Initial commit

added a,b files

added a,b files

README

techie-02

```
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ git checkout hotfix
Switched to a new branch 'hotfix'
branch 'hotfix' set up to track 'origin/hotfix'.

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git branch --list
* hotfix
  main
  release

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ |
```

Conclusion:

Successfully created and configured a local branch from the GitHub branch.

Task 8: Merge the created branch with master in git local.

Objective:

Merge a feature branch into the main branch locally.

Purpose:

To combine completed feature changes with the main development branch.

Implementation:

1. Switched to the main branch – git checkout main.

2. Pulled the latest changes – git pull.
3. Merged the feature branch – git merge hotfix.
4. Resolved conflicts if required.

```
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ touch alphabet

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git add .

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git commit -m "added alphabet file"
[hotfix e690726] added alphabet file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 alphabet

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git checkout main
Switched to branch 'main'
your branch is up to date with 'origin/main'.

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ ls
README.md  a  b

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ git merge hotfix
Updating 2dfb5d1..e690726
Fast-forward
 alphabet | 0
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 alphabet

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ ls
README.md  a  alphabet  b

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$
```

Conclusion:

Successfully merged the feature branch into the main branch locally.

Task 9: Merge the created branch with master in github by sending a pull request.

Objective:

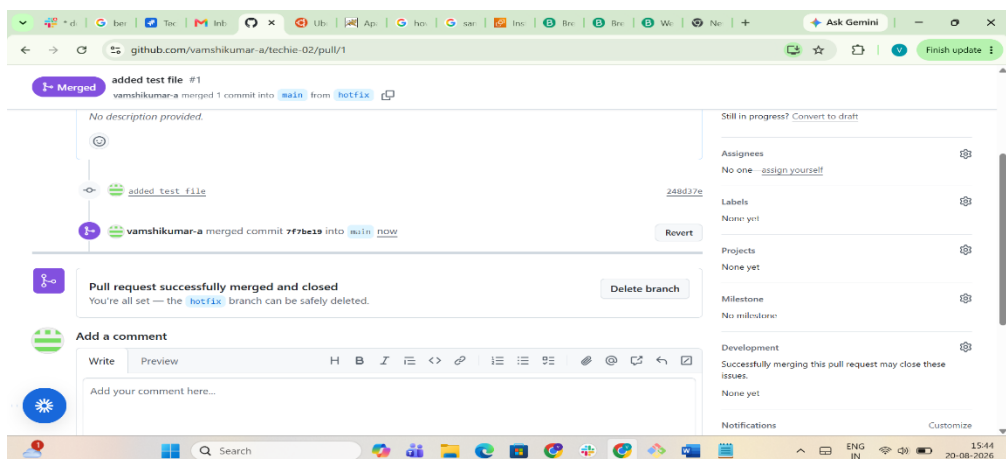
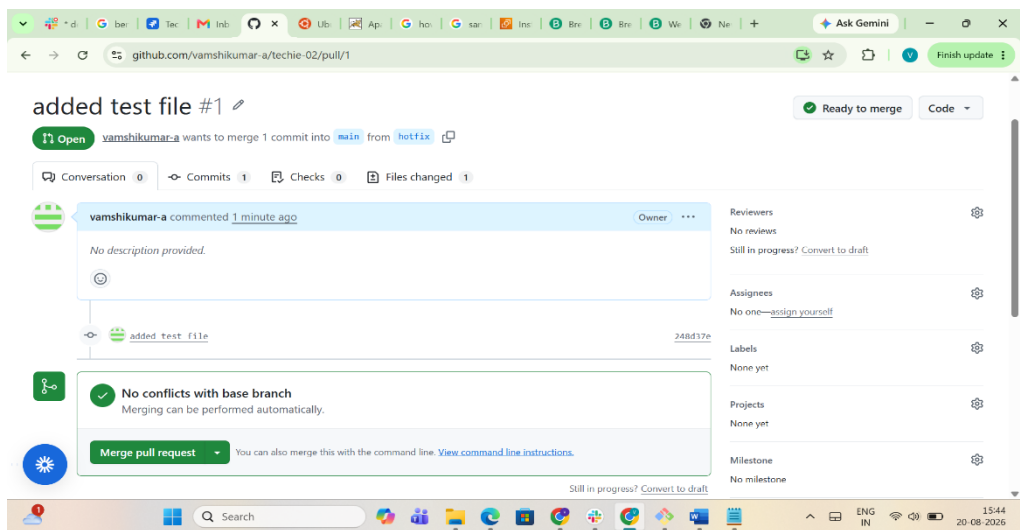
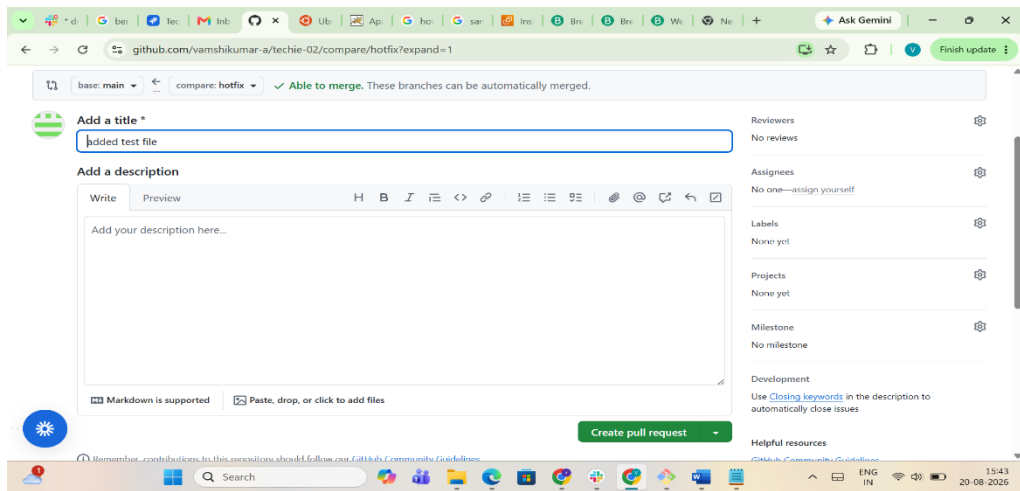
Merge a feature branch into the main branch through a Pull Request.

Purpose:

To understand the GitHub collaboration and code-review workflow.

Implementation:

1. Pushed the hotfix branch to GitHub.
2. Opened the GitHub repository.
3. Created a Pull Request.
4. Selected the hotfix branch as the source.
5. Selected main as the target.
6. Reviewed the changes.
7. Created and merged the Pull Request.



Conclusion:

Successfully merged the feature branch into main using a GitHub Pull Request.

Task 10: Create a file in local and send that to branch in github.

Objective:

Create a new file locally and push it to a specific branch.

Purpose:

To practice making branch-specific changes and synchronizing them with GitHub.

Implementation:

1. Switched to the required branch – git checkout hotfix.
2. Created a new file – touch test.
3. Added the file to staging – git add . .
4. Committed the changes – git commit -m "message".
5. Pushed the branch to GitHub – git push origin.
6. Verified the file on GitHub.

```
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ touch test

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ ls
README.md  a  alphabet  b  test

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git add .

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git commit -m "added test file"
[hotfix 248d37e] added test file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 test
```

```
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git push origin
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 260 bytes | 260.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:vamshikumar-a/techie-02.git
2dfb5d1..248d37e hotfix -> hotfix
```

Conclusion:

Successfully pushed a new file to the GitHub branch.

Task 11: Clone only a branch from github to local.

Objective:

Clone only a selected branch from a GitHub repository.

Purpose:

To work with a specific branch without cloning all remote branches.

Implementation:

1. Identified the required branch.
2. Used the -b option with git clone.
3. Used --single-branch.
4. Verified the active branch.

```
SLF3BRIDEAPAD MINGW64 ~/Desktop
$ ls
'Class half yearly question paper.docx'  'Marriage items.xlsx'  'Science_Question_Paper (1) (2).pdf'  desktop.ini
'A VAMSHI KUMAR (2).docx'                'MongoDB Compass.link' 'Science_Question_Paper (1).docx'      techie-02/
'A VAMSHI-MERN-STACK.pdf'                'Muvavi Video Editor 25.link' 'Science_Question_Paper (1).pdf'       vamsht_projects/
'ASV PDF.pdf'                            'New Microsoft Excel Worksheet (2).xlsx' 'Visual Studio Code.link'             wordpress/
'ASV-Apartment Unpriced BOQ.xlsx'         'New Microsoft Excel Worksheet.xlsx'    'Zoom Workplace.link'                 '~New Microsoft Excel Worksheet (2).xlsx'
House-plan/                               'PLY WOOD SHUTTERING QUOTATION.pdf'    desktop/

SLF3BRIDEAPAD MINGW64 ~/Desktop
$ rm -rf techie-02/

SLF3BRIDEAPAD MINGW64 ~/Desktop
$ ls
'Class half yearly question paper.docx'  'Marriage items.xlsx'  'Science_Question_Paper (1) (2).pdf'  desktop.ini
'A VAMSHI KUMAR (2).docx'                'MongoDB Compass.link' 'Science_Question_Paper (1).docx'      techie-02/
'A VAMSHI-MERN-STACK.pdf'                'Muvavi Video Editor 25.link' 'Science_Question_Paper (1).pdf'       vamsht_projects/
'ASV PDF.pdf'                            'New Microsoft Excel Worksheet (2).xlsx' 'Visual Studio Code.link'             wordpress/
'ASV-Apartment Unpriced BOQ.xlsx'         'New Microsoft Excel Worksheet.xlsx'    'Zoom Workplace.link'                 '~New Microsoft Excel Worksheet (2).xlsx'
House-plan/                               'PLY WOOD SHUTTERING QUOTATION.pdf'    desktop/

SLF3BRIDEAPAD MINGW64 ~/Desktop
$ git clone -b main git@github.com:vamshikumar-a/techie-02.git
Cloning into 'techie-02'...
remote: Enumerating objects: 14, done.
remote: Counting objects: 100% (14/14), done.
remote: Compressing objects: 100% (10/10), done.
remote: Total 14 (delta 8); reused 6 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (14/14), done.
Resolving deltas: 100% (6/6), done.

SLF3BRIDEAPAD MINGW64 ~/Desktop
$ cd techie-02

SLF3BRIDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ git branch --list
* main

SLF3BRIDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$
```

Conclusion:

Successfully cloned only the required branch.

Task 12: Create a file with all passwords and make that untrackable with git.

Objective:

Prevent a sensitive password file from being tracked by Git.

Purpose:

To avoid accidentally committing sensitive information to the repository.

Implementation:

1. Created a .env file.
2. Added the file to .gitignore.
3. Checked the Git status.

```

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ ls -a
./ ../ .git/ .gitignore README.md a alphabet b test

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ vi .gitignore

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ touch .env

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ vi .env

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)
$ ls -a
./ ../ .env .git/ .gitignore README.md a alphabet b test

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (main)

```

Conclusion:

Successfully configured Git to ignore the password file.

Task 13: Make a commit and make that commit reset without savings changes.

Objective:

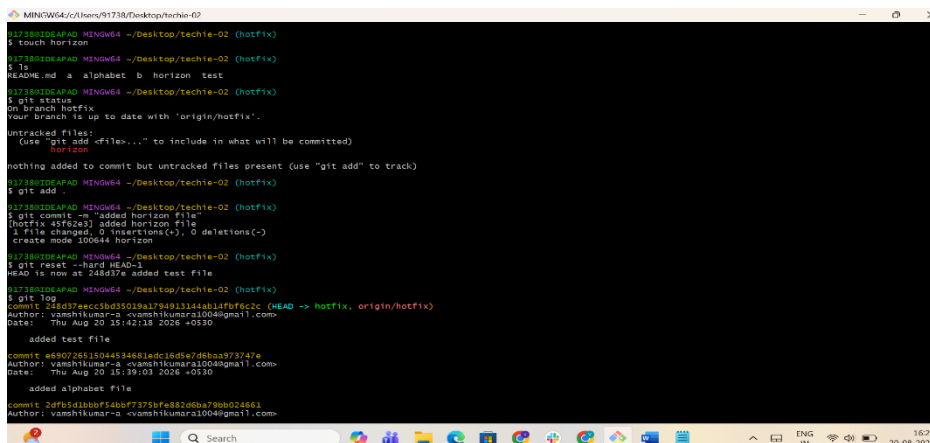
Remove the latest commit and discard its changes.

Purpose:

To understand how git reset --hard can remove a commit and its changes.

Implementation:

1. Created file – touch horizon.
2. Commit changes – git commit -m “message”.
3. Used git reset --hard HEAD~1.
4. Verified that the commit and changes were removed.



```

MINGW64~/Desktop/techie-02
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ touch horizon

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ ls
README.md a alphabet b horizon test

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git status
On branch hotfix
Your branch is up to date with 'origin/hotfix'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    horizon

nothing added to commit but untracked files present (use "git add" to track)

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git add .

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git commit -m "added horizon file"
[hotfix:4f8a2a3] added horizon file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 horizon

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git reset --hard HEAD~1
HEAD is now at 24b337e added test file

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git log
commit 24b337eccc7bd35019a1794913144a14fbfec2c (HEAD -> hotfix, origin/hotfix)
Author: vamsaikumar-a <vamsaikumar1004@gmail.com>
Date: Thu Aug 20 15:42:18 2026 +0530
    added test file

commit a89072613044534661edc16de7d6ba373747e
Author: vamsaikumar-a <vamsaikumar1004@gmail.com>
Date: Thu Aug 20 15:39:03 2026 +0530
    added alphabet file

commit 24b337eccc7bd35019a1794913144a14fbfec2c
Author: vamsaikumar-a <vamsaikumar1004@gmail.com>

```

Conclusion:

Successfully reset the commit and discarded its changes.

Task 14: Revert a committed commit to the older version.

Objective:

Undo the changes made by a previous commit.

Purpose:

To reverse changes while keeping the existing Git history.

Implementation:

1. Viewed the commit history.
2. Identified the required commit ID.
3. Used git revert.
4. Verified that a new revert commit was created.
5. Pushed the changes to GitHub.

```
U1738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git log
commit 248d37eccc5bd35019a1794913144ab14fbf6c2c (HEAD -> hotfix, origin/hotfix)
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Thu Aug 20 15:42:18 2026 +0530

    added test file

commit e690726515044534681edc16d5e7d6baa973747e
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Thu Aug 20 15:39:03 2026 +0530

    added alphabet file

commit 2dfb5d1bbbf54bbf7375bfe882d6ba79bb024661
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Thu Aug 20 15:27:01 2026 +0530

    added a,b files

commit 9cd8c92de4890c846f9b5227427c979f1c1251ad
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Thu Aug 20 15:21:45 2026 +0530

    Initial commit

U1738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ ls
README.md a alphabet b test

U1738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git revert 2dfb5d
[hotfix 221d608] Revert "added a,b files"
 2 files changed, 0 insertions(+), 0 deletions(-)
 delete mode 100644 a
 delete mode 100644 b

U1738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ ls
README.md alphabet test
```

Conclusion:

Successfully reverted a previous commit while preserving Git history.

Task 15: Push a file to stash without saving the changes and work on another file.

Objective:

Temporarily save unfinished changes and work on another task.

Purpose:

To switch tasks without committing incomplete work.

Implementation:

1. Modified a file.
2. Checked the repository status.
3. Used git stash.
4. Verified that the working directory was clean.
5. Worked on another file.

```
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ touch a

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git stash
No local changes to save

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git add .

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git stash
Saved working directory and index state WIP on hotfix: 221d608 Revert "added a,b files"
```

```
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git stash list
stash@{0}: WIP on hotfix: 221d608 Revert "added a,b files"

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git status
On branch hotfix
Your branch is up to date with 'origin/hotfix'.

nothing to commit, working tree clean

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ touch file2.txt

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git add .

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git commit -m "worked on file2"
[hotfix beb67e3] worked on file2
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 file2.txt

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git pull
Already up to date.

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git push origin
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
```

Conclusion:

Successfully stashed unfinished changes and continued working on another file.

Task 16: Undo the stash file and start working on that again.

Objective:

Restore previously stashed changes.

Purpose:

To retrieve unfinished work and continue development.

Implementation:

1. Checked the stash list.
2. Identified the required stash.

3. Used git stash pop.

4. Verified that the changes were restored.

```
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git stash pop
On branch hotfix
Your branch is up to date with 'origin/hotfix'.

Changes to be committed:
  (use "git restore --staged <File>..." to unstage)
    new file:   a

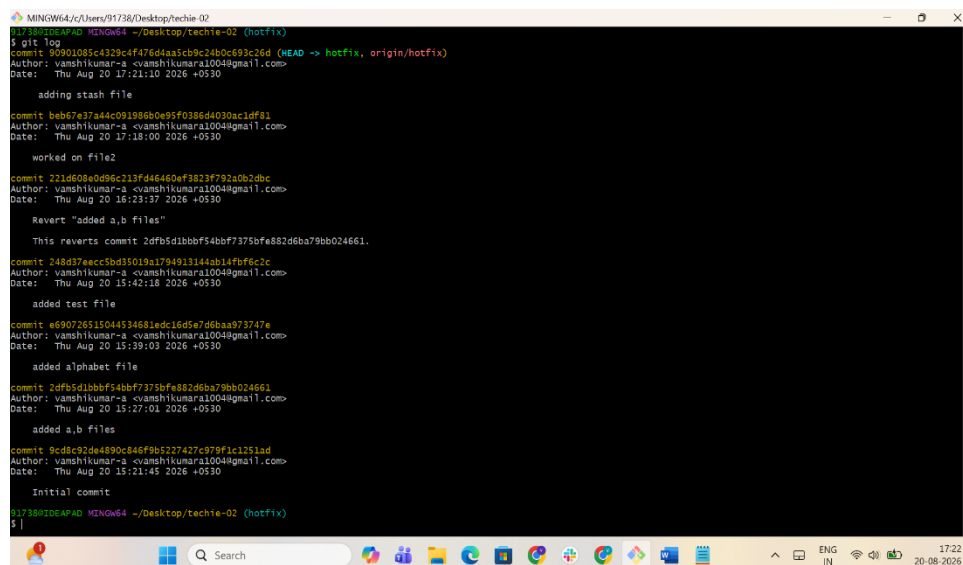
Dropped refs/stash@{0} (651ba85bf35b0594427e109615ba4d27dbbfde71)

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ ls
README.md  a  alphabet  file2.txt  test
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ vi a
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git add .
warning: in the working copy of 'a', LF will be replaced by CRLF the next time Git touches it
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git status
On branch hotfix
Your branch is up to date with 'origin/hotfix'.

Changes to be committed:
  (use "git restore --staged <File>..." to unstage)
    new file:   a

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git commit -m "adding stash file"
[hotfix 9090108] adding stash file
1 file changed, 2 insertions(+)
create mode 100644 a

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git push origin
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 275 bytes | 275.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
```



```
MINGW64/c/Users/91738/Desktop/techie-02
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git log
commit 90901085c4329c4f476d4a5cb9c24b0c693c28d (HEAD -> hotfix, origin/hotfix)
Author: vamsnikumar-a <vamsnikumara1004@gmail.com>
Date: Thu Aug 20 17:21:10 2026 +0530

    adding stash file

commit beb67e37a44c091986b0e95f0386d4030ac1df81
Author: vamsnikumar-a <vamsnikumara1004@gmail.com>
Date: Thu Aug 20 17:28:00 2026 +0530

    worked on file2

commit 221d608e0d96c213fd46460ef3823f792a0b2dec
Author: vamsnikumar-a <vamsnikumara1004@gmail.com>
Date: Thu Aug 20 16:23:37 2026 +0530

    Revert "added a,b files"

    This reverts commit 2dfb5d1bbbf54bbf7375bfe882d6ba79bb024661.

commit 248d37eccc5bd35019a1794913144ab14fbf6c2c
Author: vamsnikumar-a <vamsnikumara1004@gmail.com>
Date: Thu Aug 20 15:42:16 2026 +0530

    added test file

commit e690726515044534681edc16d5e7d6baa973747e
Author: vamsnikumar-a <vamsnikumara1004@gmail.com>
Date: Thu Aug 20 15:39:03 2026 +0530

    added alphabet file

commit 2dfb5d1bbbf54bbf7375bfe882d6ba79bb024661
Author: vamsnikumar-a <vamsnikumara1004@gmail.com>
Date: Thu Aug 20 15:27:01 2026 +0530

    added a,b files

commit 9cd8c92de4890c846f9b5227427c979f1c1251ad
Author: vamsnikumar-a <vamsnikumara1004@gmail.com>
Date: Thu Aug 20 15:21:45 2026 +0530

    Initial commit

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$
```

Conclusion:

Successfully restored the stashed changes and continued working.

Task 17: Generate a ssh-keygen and configure into github.

Objective:

Configure SSH authentication between the local system and GitHub.

Purpose:

To securely connect to GitHub without entering credentials for every Git operation.

Implementation:

1. Generated an SSH key pair.

2. Displayed the public key.
3. Copied the public key.
4. Added the key to GitHub under **Settings** → **SSH and GPG keys**.

```
91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git remote add origin git@github.com:vamshikumar-a/techie.git

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git push -u origin master
The authenticity of host 'github.com (20.207.73.82)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvV6TuJJhbpZisF/zLDA0zPMSvHdKr4UvCOqU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
git@github.com: Permission denied (publickey).
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/c/Users/91738/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/91738/.ssh/id_rsa
Your public key has been saved in /c/Users/91738/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:kd+sF6QBqRmRyisijgxSvVaj8ys9KMMSKCKgXhCJEZ0 91738@IDEAPAD
The key's randomart image is:
+---[RSA 3072]-----+
|++o+..
|.oE ..
|...+ +.
|oo+ B
|+oo .S
|#... .o
|/*.. ..
|X=* .
|o*o+
+---[SHA256]-----+

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ cat /c/Users/91738/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQBgQC5IE0T+Z4wu9FaGiX04Gf7hLKdrM0qpqqXrU8Fes0NxJ9tWZJ7IhhDAH2VzVjQmUdK86
```

Last used within the last week — Read/write



local-pc

SHA256:kd+sF6QBqRmRyisijgxSvVaj8ys9KMMSKCKgXhCJEZ0

Added on Aug 18, 2026

SSH

Last used within the last week — Read/write

Delete

Check out our guide to [connecting to GitHub using SSH keys](#) or troubleshoot [common SSH problems](#).

GPG keys

New GPG key

There are no GPG keys associated with your account

```
91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$ git remote -v
origin  git@github.com:vamshikumar-a/techie.git (fetch)
origin  git@github.com:vamshikumar-a/techie.git (push)

91738@IDEAPAD MINGW64 ~/Desktop/hiring-app (master)
$
```

Conclusion:

Successfully generated and configured an SSH key for GitHub authentication.

Task 18: Configure webhooks to github.

Objective:

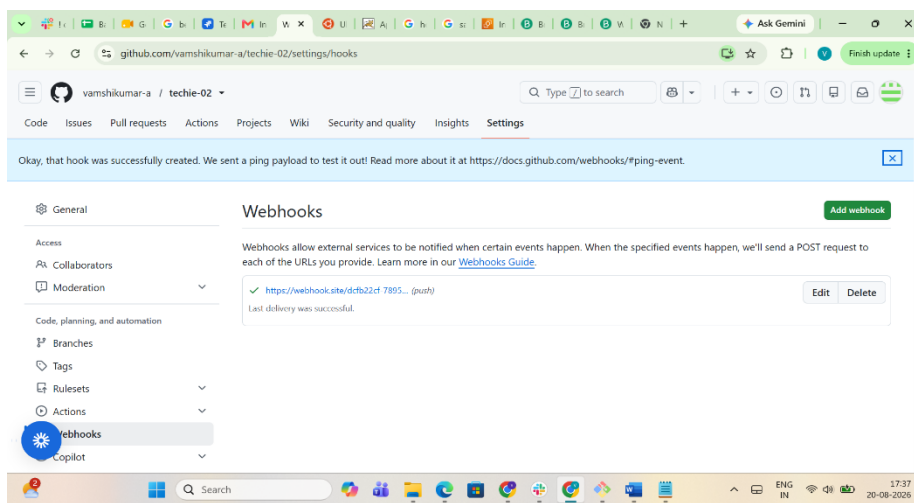
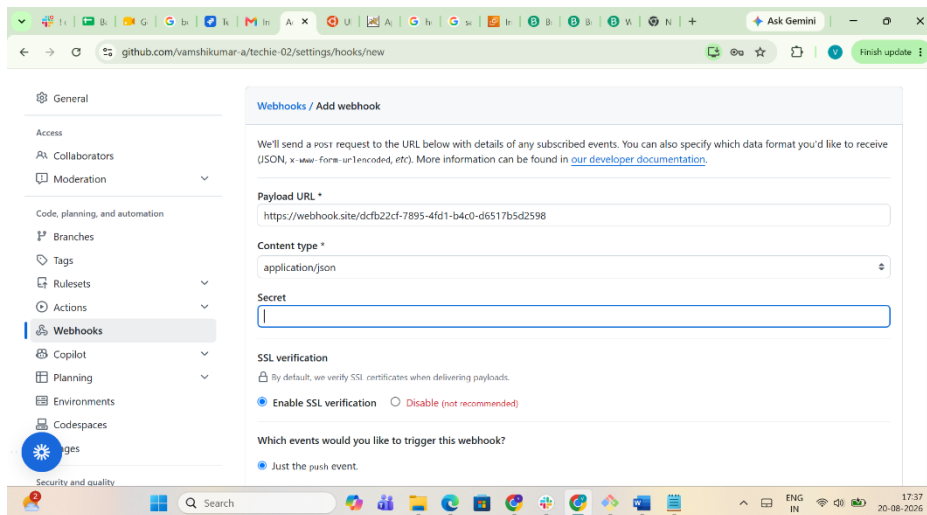
Configure a webhook to receive GitHub repository events.

Purpose:

To understand how GitHub automatically sends repository events to an external application.

Implementation:

1. Created a webhook testing URL using Webhook.site.
2. Opened the GitHub repository.
3. Navigated to **Settings** → **Webhooks**.
4. Clicked **Add webhook**.
5. Added the Payload URL.
6. Selected application/json.
7. Selected the **Push event**.
8. Enabled SSL verification.
9. Enabled the webhook.
10. Made a Git push to test the webhook.
11. Verified the webhook delivery.



```
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ ls
README.md a alphabet file2.txt test webhook-test.txt

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git status
On branch hotfix
Your branch and 'origin/hotfix' have diverged,
and have 1 and 4 different commits each, respectively.
(use "git pull" to merge the remote branch into yours)

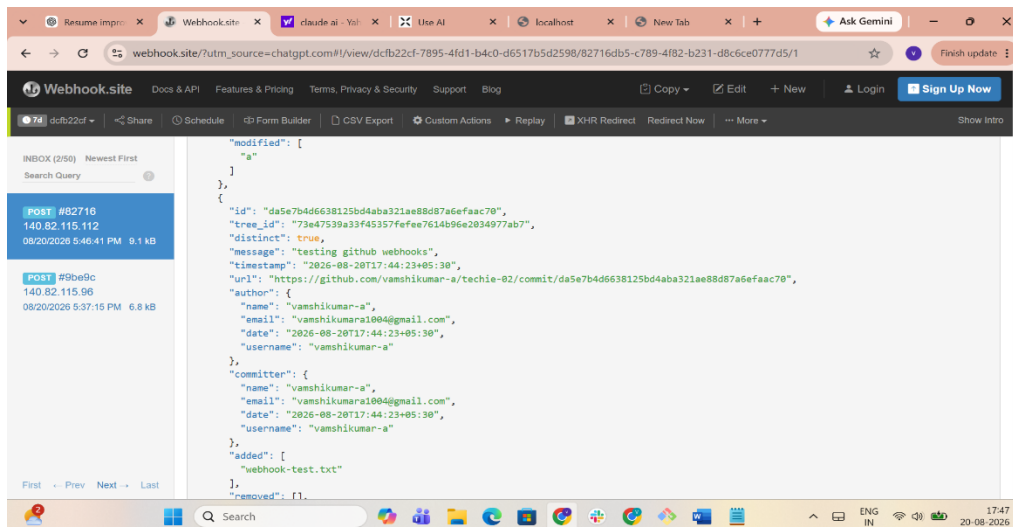
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        webhook-test.txt

nothing added to commit but untracked files present (use "git add" to track)

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git add .
warning: in the working copy of 'webhook-test.txt', LF will be replaced by CRLF the next time Git touches it

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git commit -m "testing github webhooks"
[hotfix da5e7b4] testing github webhooks
1 file changed, 1 insertion(+)
create mode 100644 webhook-test.txt

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
```



Conclusion:

Successfully configured and tested a GitHub webhook.

Task 19: Basic understanding of .git file.

Objective:

Understand the basic structure of the .git directory.

Purpose:

To understand where Git stores repository metadata, configuration, branches, and history.

Implementation:

1. Initialized a Git repository.
2. Located the .git directory.
3. Listed its contents.
4. Reviewed important files and directories such as HEAD, config, objects, refs, index, and logs.

```
91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ cd .git/

91738@IDEAPAD MINGW64 ~/Desktop/techie-02/.git (GIT_DIR!)
$ ls
COMMIT_EDITMSG  FETCH_HEAD  HEAD  ORIG_HEAD  REBASE_HEAD  config  description  hooks/  index  info/  logs/  objects/  packed-refs  refs/
```

Conclusion:

Successfully reviewed the basic structure and purpose of the .git directory.

Task 20: Check all the logs of git.

Objective:

View the complete commit history of the repository.

Purpose:

To track previous commits, changes, branches, and authors.

Implementation:

1. Used git log to view commit history.

```
91738010EAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git log
commit 8aa40de207f173a56da7923261bba368b0d568e7 (HEAD -> hotfix)
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Thu Aug 20 17:21:10 2026 +0530

    updating the stash file

commit beb67e37a44c091986b0e95f0386d4030ac1df81
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Thu Aug 20 17:18:00 2026 +0530

    worked on file2

commit 221d608e0d96c213fd46460ef3823f792a0b2dbc
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Thu Aug 20 16:23:37 2026 +0530

    Revert "added a,b files"

    This reverts commit 2dfb5d1bbbf54bbf7375bfe882d6ba79bb024661.

commit 248d37eccc5bd35019a1794913144ab14fbf6c2c
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Thu Aug 20 15:42:18 2026 +0530

    added test file

commit e690726515044534681edc16d5e7d6baa973747e
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Thu Aug 20 15:39:03 2026 +0530

    added alphabet file

commit 2dfb5d1bbbf54bbf7375bfe882d6ba79bb024661
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Thu Aug 20 15:27:01 2026 +0530

    added a,b files

commit 9cd8c92de4890c846f9b5227427c979f1c1251ad
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Thu Aug 20 15:21:45 2026 +0530

    Initial commit
```

Conclusion:

Successfully viewed and analyzed the Git commit history.

Task 21: Rename the commit message.

Objective:

Modify an existing commit message.

Purpose:

To maintain clear and meaningful commit messages.

Implementation:

1. Checked the current commit history.
2. Identified the latest commit.
3. Used git commit --amend.
4. Updated the commit message.
5. Verified the updated history.

```

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git commit --amend
[hotfix 8aa40de] updating the stash file
Date: Thu Aug 20 17:21:10 2026 +0530
1 file changed, 2 insertions(+)
create mode 100644 a

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git log
commit 8aa40de207f173a56da7923261bba368b0d568e7 (HEAD -> hotfix)
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Thu Aug 20 17:21:10 2026 +0530

    updating the stash file

commit beb67e37a44c091986b0e95f0386d4030ac1df81
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Thu Aug 20 17:18:00 2026 +0530

    worked on file2

commit 221d608e0d96c213fd46460ef3823f792a0b2dbc

```

Conclusion:

Successfully renamed the latest commit message.

Task 22: Merge multiple commits into single commit.

Objective:

Combine multiple commits into a single commit.

Purpose:

To maintain a clean and organized Git history.

Implementation:

1. Checked the commit history.
2. Identified the commits to combine.
3. Started an interactive rebase.
4. Kept the first commit as pick.
5. Changed the remaining commits to squash.
6. Updated the final commit message.
7. Verified the updated history.

```

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git rebase -i HEAD~4
[detached HEAD e56f493] added test file
Date: Thu Aug 20 15:42:18 2026 +0530
3 files changed, 2 insertions(+)
rename b => file2.txt (100%)
create mode 100644 test
Successfully rebased and updated refs/heads/hotfix.

91738@IDEAPAD MINGW64 ~/Desktop/techie-02 (hotfix)
$ git log
commit e56f493d181a680f8d5a36eeeb45c245c0580cb (HEAD -> hotfix)
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Thu Aug 20 15:42:18 2026 +0530

    added test file

Revert "added a,b files"

This reverts commit 2dfb5d1bbbf54bbf7375bfe882d6ba79bb024661.

worked on file2

updating the stash file

commit e690726515044534681edc16d5e7d6baa973747e
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Thu Aug 20 15:39:03 2026 +0530

    added alphabet file

commit 2dfb5d1bbbf54bbf7375bfe882d6ba79bb024661
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Thu Aug 20 15:27:01 2026 +0530

    added a,b files

commit 9cd8c92de4890c846f9b5227427c979f1c1251ad
Author: vamshikumar-a <vamshikumara1004@gmail.com>
Date: Thu Aug 20 15:21:45 2026 +0530

    Initial commit

```

Conclusion:

Successfully combined multiple commits into a single commit using interactive rebase.